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ORIGINAL ARTICLE

Web tracking analysis on top websites through OpenWPM analytics enhancements

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Abstract

Nowadays, websites employ multiple tracking mechanisms to personalize user experiences and deliver targeted advertisements. While cookies remain the dominant tracking tool, recent trends indicate an increasing adoption of stateless fingerprinting based on JavaScript methods. However, existing studies often focus on specific tracking technologies, while detailed analyses that jointly examine multiple stateful and stateless tracking mechanisms remain limited. To address this gap, we enhanced OpenWPM-Data-Analytics to improve the detection of fingerprinting mechanisms in scans performed with OpenWPM, a widely used framework for web privacy measurement studies. Using this enhanced measurement and analysis pipeline, we conducted an empirical analysis of the top 1000 most-visited websites, jointly examining cookies, web storage, web analytics tools, and browser fingerprinting techniques. The results show that stateful tracking methods such as cookies and web storage remain highly prevalent. However, fingerprinting techniques were detected on nearly one third of the analyzed websites, with JavaScript Enumeration and WebGL being the most common mechanisms. This prevalence is higher than that reported in previous automated studies, suggesting a growing reliance on fingerprinting as third-party cookies face increasing restrictions. In addition, web analytics tools were present in half of the websites analyzed. These findings highlight the growing complexity of modern web tracking and the need to improve current detection and mitigation strategies, particularly against fingerprinting mechanisms that are more difficult to block and regulate.

Keywords Web tracking · Fingerprinting · Stateful web tracking · Stateless web tracking · Privacy assessment · OpenWPM · Privacy

1 Introduction

Modern websites rely heavily on user tracking technologies to support personalization, analytics, fraud prevention, and the delivery of targeted advertising [34, 44]. These tracking techniques have evolved from simple, stateful methods such as cookies to sophisticated stateless approaches like browser fingerprinting [10, 17]. This evolution has made tracking harder to detect, regulate, and block, thereby intensifying privacy concerns.

At the same time, different regulatory frameworks have been introduced to increase transparency and give users more control over their data [14, 50, 54]. However, the rapid evolution of tracking technologies continues to challenge effective enforcement and privacy protection [7, 13, 17, 46, 56, 61]. In particular, many fingerprinting techniques operate without users' awareness and can bypass traditional consent mechanisms.

Despite extensive research on web tracking, existing studies often focus on specific tracking technologies or limited subsets of tracking mechanisms. Although some studies consider multiple tracking technologies [10, 27], detailed analyses jointly examining different stateful and stateless tracking mechanisms remain limited. This fragmentation makes it difficult to obtain a comprehensive view of how different tracking technologies coexist on modern websites and to assess their evolution, particularly as restrictions on traditional mechanisms such as third-party cookies increase.

Most large-scale web privacy studies rely on privacy scanners, i.e., automated crawlers designed to detect tracking technologies across websites [7, 10, 17]. These tools enable scalable and reproducible measurements over thousands of

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websites at low operational cost and without exposing real users to privacy risks. Consequently, they have become a standard methodology in web privacy research.

However, privacy scanners also have important limitations, particularly for detecting fingerprinting behaviors triggered only after user interactions. Annamalai et al. [46], for instance, showed that 45% of fingerprinting websites observed during real-user browsing sessions were entirely missed by privacy scanners, suggesting that crawler-based measurements may underestimate fingerprinting prevalence.

Nevertheless, privacy scanners remain essential for large-scale and reproducible web privacy measurements. Frameworks such as Open Web Privacy Measurement (OpenWPM) [17] and FPDetective [1] provide browser instrumentation that enables the collection of data associated with different tracking behaviors. Motivated by the need for a more comprehensive view of modern web tracking, this work investigates the extent to which multiple stateful and stateless tracking mechanisms coexist across highly visited websites. To this end, this paper makes the following two main contributions.

First, we conduct *a large?scale and detailed measurement study of multiple tracking mechanisms* across the Tranco Top?1000 sites, jointly analyzing cookies, web storage, web analytics tools, and browser fingerprinting techniques. This analysis provides an updated view of the prevalence and coexistence of multiple stateful and stateless tracking mechanisms on highly visited websites, with the observed fingerprinting prevalence suggesting a growing adoption of these techniques compared with previous automated studies.

Second, to support this measurement study, we *extend OpenWPM's JavaScript instrumentation and enhance OpenWPM?Data?Analytics*, a post-processing framework *built on top of OpenWPM*. These enhancements extend the measurement and analysis pipeline to support the detection and analysis of the tracking mechanisms considered in this work. The resulting enhancements are open source and designed to preserve compatibility with OpenWPM-based measurement workflows, facilitating reproducibility and longitudinal studies.

Unlike fingerprinting detection approaches such as FP-Inspector [30] or FP-Tracer [10], which rely on heavier browser instrumentation, semantic analysis, or dynamic taint tracking, our proposal relies on lightweight instrumentation and post-processing of OpenWPM traces. This design choice prioritizes scalability, reproducibility, and compatibility with OpenWPM-based measurement workflows.

The rest of the article is organized as follows. Section 2 provides background regarding tracking mechanisms and privacy scanners. Section 3 presents related work. Section 4 describes the methodology followed in this research. Section 5 explains how different web tracking mechanisms can be detected using OpenWPM and describes the enhancements introduced to the OpenWPM-based measurement and analysis pipeline. Using these tools, Sect. 6 presents an analysis of the tracking mechanisms used by the top websites. Section 7 discusses these results. Finally, Sect. 8 presents the main conclusions and future work.

2 Background

Web tracking mechanisms can be broadly distinguished according to whether user identification relies on information stored on the client side or on characteristics obtained from the user's browser and device [16, 23, 61]. In this work, we consider both stateful tracking mechanisms, including cookies and web storage, and stateless mechanisms based on browser fingerprinting. We also consider web analytics tools, which may rely on different tracking mechanisms to collect information about website usage and user behavior.

2.1 Stateful tracking mechanisms

Stateful tracking relies on information stored on the client side that can be retrieved in subsequent interactions. Among the mechanisms considered in this work are cookies and web storage.

2.1.1 Cookies

Cookies are the mechanism through which an HTTP connection can maintain state. They are established by the server via the *Set-Cookie* header and returned by the client through the *Cookie* header. By default, their lifetime lasts until the end of the session, although the server may define a specific expiration time. Their functionality, security, and implementation considerations are described in RFC 6265 [9]. This document also specifies the attributes introduced to improve cookie security and privacy.

Depending on the relationship between the domain setting the cookie and the website being visited, cookies can be classified as first-party or third-party cookies. First-party cookies are associated with the origin website, whereas third-party cookies are associated with external domains included by the visited website.

2.1.2 Web storage

Web storage allows arbitrary information to be stored in the form of key-value entries. It can be either local, where the stored data persists across browser sessions, or session-based, where the information is deleted when the session ends—although some browsers may preserve it. Furthermore, IndexedDB [45] stores information in a structured way similar to NoSQL databases, such as MongoDB, and may

persist the stored data indefinitely, similarly to local storage. Thus, a website may store identifiers on the client side, either during a session or until the stored data is manually removed.

2.2 Stateless tracking: browser fingerprinting

Browser fingerprinting emerged as an alternative to tracking users through cookies or web storage [15, 39]. Browser fingerprinting can be classified into passive and active techniques [59]. Passive fingerprinting relies on information that servers can gather from HTTP headers. In contrast, active fingerprinting leverages JavaScript APIs and HTML5 objects to extract unique device and browser characteristics, allowing websites to create browser or device fingerprints without relying on locally stored identifiers. Browser fingerprinting is used for targeted advertising, user authentication, usability analysis, and fraud detection [21, 38, 53].

Initially, fingerprinting relied on HTTP headers and a limited number of APIs [17]. Over time, additional mechanisms such as Canvas, WebGL, fonts, extensions, and APIs related to audio, battery status, performance, and sensors have come into use [7]. This evolution has increased the variety of browser and device characteristics that can be exploited for fingerprinting.

This work focuses on active browser fingerprinting techniques that can be observed through browser instrumentation. Specifically, the study considers Audio Context, Canvas, Canvas Font, WebGL, WebRTC, and JavaScript (JS) Enumeration. The specific conditions used to detect each technique are described later as part of the measurement methodology.

2.3 Web analytics tools

Web analytics tools collect, analyze, and report information about user activity and behavior when visiting a website, typically with the aim of optimizing the user experience. To achieve this, they rely on multiple tracking mechanisms to gather the required data and build visitor profiles. In some cases, these profiles may be stored for internal use or shared across different customers and services.

In this study, web analytics tools are considered separately from the underlying tracking mechanisms they may employ. The specific analytics platforms included in the study and the criteria used for their detection are described later as part of the measurement methodology.

2.4 Privacy scanners and OpenWPM

Privacy scanners enable automated and reproducible measurements of tracking mechanisms across large sets of websites. These tools typically automate browser visits while collecting information about network requests, cookies, storage accesses, and JavaScript execution, allowing researchers to analyze tracking behaviors at scale [7, 10, 17].

Open Web Privacy Measurement (OpenWPM) [17] is an open-source web privacy measurement framework built around automated browser instrumentation. It provides mechanisms for collecting browsing data, including HTTP requests and responses, cookies, and JavaScript API accesses, which can subsequently be analyzed to identify privacy-related behaviors. Its extensible instrumentation and structured data collection have made it suitable for large-scale studies of web tracking.

In this work, OpenWPM provides the underlying browser instrumentation and data collection framework. The specific configuration used in our measurements, the extensions introduced to its JavaScript instrumentation, and the subsequent analysis performed using OpenWPM-Data-Analytics are described in the methodology (Sect. 4).

3 Related work

Previous research has investigated web tracking from different perspectives, ranging from traditional stateful mechanisms such as cookies and web storage to stateless techniques based on browser fingerprinting [7, 17, 27, 46, 49, 59]. However, these studies differ considerably in the tracking mechanisms considered, the website populations analyzed, and the measurement and detection methodologies employed. In this section, we therefore focus on comparing the scope of previous measurement studies and their coverage of the tracking mechanisms considered in this work.

Several studies have focused on stateful tracking mechanisms, particularly cookies and web storage [13, 24, 49, 61]. Englehardt and Narayanan [17] provided an early large-scale characterization of cookie-based tracking, while more recent research has investigated how cookie tracking is changing as browsers and privacy protections increasingly restrict third-party cookies. Lin et al. [37], for example, studied tracking in a cookieless context, whereas Chen et al. [11] showed how trackers can increasingly rely on first-party cookies. Web storage has also received attention as an alternative client-side mechanism. Ahmad et al. [2], for instance, investigated the use of different web storage technologies across 10,000 Tranco websites. These works provide detailed analyses of particular stateful mechanisms, but generally do not combine them with a broad analysis of fingerprinting and web analytics within the same measurement setting.

Other studies have examined how consent interfaces and browsing conditions influence cookie deployment. Balleys et al. [8] analyzed "accept-or-pay" cookie banners, while research on dark patterns has shown that interface design can influence users' decisions regarding tracking [26, 28, 48, 51, 58]. Goanta et al. [24] further showed that user interaction

Table 1 Tracking detection mechanisms supported by the different web privacy scanners

Tools	HTTP headers	Cookies	Web storage			Active fingerprinting					Web analytics tools			
			Local storage	Session storage	IndexedDB	Audio context	Canvas	Canvas font	WebGL	WebRTC	JS enumeration	Google analytics	Facebook pixel	Microsoft UET	Hotjar
OpenWPM	✓	✓	✓	✓	×	✓	✓	×	✓	✓	△	×	×	×	×
PrivacyScore	✓	✓	✓	×	×	×	×	×	×	×	×	✓	×	×	×
PrivacyScanner	✓	✓	×	×	×	×	✓	×	×	×	×	✓	×	×	×
iPrivacyScanner	✓	✓	✓	✓	×	×	✓	×	✓	✓	✓	✓	✓	×	×
WebbKoll	✓	✓	✓	✓	×	×	×	×	×	×	×	×	×	×	×
Blacklight	✓	✓	×	×	×	×	✓	✓	×	×	×	✓	✓	×	×
FP-Inspector	×	×	×	×	×	✓	✓	✓	✓	✓	✓	×	×	×	×
FP-Tracer	✓	×	✓	✓	×	✓	✓	✓	✓	✓	✓	×	×	×	×
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notation: ✓ = Supported, × = Not supported, △ = Partially supported; partial support indicates that additional instrumentation and/or post-processing is required for the complete detection criterion used in this study

with consent banners, geographic location, and device type affect the number and type of cookies observed. These results are also relevant from a measurement perspective, since the tracking behavior observed by automated crawlers may differ from that experienced during real-user browsing.

Fingerprinting has increasingly become the focus of dedicated measurement studies. Bahrami et al. [7] conducted a longitudinal study covering the period from 2010 to 2019 and reported an increase in fingerprinting scripts from 1.16% in 2016 to 3.70% in 2018. More recent research has continued to investigate fingerprinting as browser vendors introduce restrictions on traditional tracking mechanisms and privacy protections evolve [14, 21, 27, 46].

The breadth of the fingerprinting techniques considered also varies across studies. Pan and Ruiz-Martínez [49] analyzed several active fingerprinting mechanisms, including Canvas, WebRTC, and WebGL, while other studies have focused on the overall prevalence of fingerprinting or on particular techniques [21, 27, 46]. Other works, such as Estrada [19] and Ruiz-Martínez [55], investigated the resilience of tracking techniques against browser privacy protections and anti-tracking countermeasures. Consequently, although fingerprinting has been extensively investigated, the mechanisms and APIs included in each measurement study differ substantially.

The measurement methodology can also affect the fingerprinting behaviors observed. Annamalai et al. [46] showed that 45% of fingerprinting websites observed during real-user browsing sessions were entirely missed by privacy scanners. Their results highlight the limitations of crawler-based measurements for detecting fingerprinting behaviors that are activated only after user interactions.

A complementary line of research has focused specifically on improving fingerprinting detection [10, 30, 57]. EssentialFP [57] detects and labels sensitive information used for fingerprinting through a precomputed API fingerprint. FP-Inspector [30] follows a machine-learning-based syntactic-semantic approach, whereas FP-Tracer [10] uses dynamic taint tracking to improve fingerprinting detection. These approaches provide specialized fingerprinting detection capabilities, but their objective differs from measurement studies aimed at jointly characterizing multiple tracking mechanisms.

Privacy scanners and browser-instrumentation frameworks have enabled many of these studies to perform automated measurements at scale. OpenWPM [17] has been widely used in web privacy measurement studies [47], while PrivacyScanner and its improved version, iPrivacyScanner [49], among other tools, have been used to analyze different subsets of tracking technologies. Previous research demonstrates the value of these frameworks for large-scale and reproducible measurements [7, 17, 49], while also iden-

tifying limitations in their coverage of advanced tracking behaviors [30, 46, 49, 50].

Table 1 compares the tracking-related capabilities provided by different privacy scanners. The comparison shows that their coverage varies across tracking mechanisms and that no single scanner provides equivalent support for all the considered categories. OpenWPM provides extensive browser instrumentation and data collection capabilities, but its support for collecting and subsequently analyzing some of the tracking mechanisms considered in this work is limited. In particular, OpenWPM is marked as partially supporting JavaScript Enumeration because its default instrumentation captures only part of the API-access information required by the detection criterion used in this study, while additional instrumentation and post-processing are necessary to perform the complete analysis. For mechanisms already supported by OpenWPM, such as Canvas, WebGL, and WebRTC, our modifications refine or extend the collected JavaScript traces required by the detection criteria rather than introducing support for these mechanisms from scratch. The row "Ours" represents the complete measurement and analysis pipeline used in this study, combining the existing capabilities of OpenWPM with our extensions to its JavaScript instrumentation and the enhanced detection and analysis capabilities of OpenWPM-Data-Analytics. This comparison therefore motivates the technical extensions described in the methodology.

The capabilities of privacy scanners, however, do not necessarily correspond to the scope of the measurement studies that use them. To complement the tool-level comparison in Table 1, Table 2 summarizes the tracking mechanisms actually analyzed by the studies later considered in our quantitative comparison. Rather than reporting their prevalence values at this stage, the table indicates whether each study analyzed the corresponding mechanism. Canvas and Canvas Font are grouped under Canvas, while information related only to dataset size is omitted, as it does not correspond to a tracking mechanism. The general Fingerprinting column indicates studies that report fingerprinting as an aggregate category even when results for individual fingerprinting techniques are not provided.

As shown in Table 2, previous studies differ considerably in the breadth and granularity of the tracking mechanisms they examine. Several works concentrate on individual technologies or report fingerprinting only as an aggregate category, whereas a smaller number consider multiple stateful and stateless mechanisms within the same study [17, 27, 49]. Even among studies with broader coverage, the specific fingerprinting techniques analyzed vary substantially. Detailed analyses jointly examining cookies, web storage, web analytics, and a broad set of active browser fingerprinting techniques therefore remain limited. Our work addresses this gap by analyzing these mechanisms within the same measurement setting and through a common OpenWPM-based collection and analysis pipeline, enabling both their individual prevalence and their coexistence across highly visited websites to be examined.

From a technical perspective, our work builds on OpenWPM and OpenWPM-Data-Analytics to support this broader measurement scope. In contrast to specialized fingerprinting detection approaches, our objective is not to introduce a new standalone privacy scanner, but to extend an established OpenWPM-based measurement workflow. Specifically, we extend OpenWPM's JavaScript instrumentation to collect information required for the tracking mechanisms considered in this study and enhance OpenWPM-Data-Analytics to support their subsequent detection and analysis. These extensions address limitations identified in the tool-level comparison above and enable the joint measurement of the tracking mechanisms considered in this work.

4 Methodology

The methodology adopted in this work follows a structured and reproducible workflow designed to detect, measure, and quantify multiple types of web tracking mechanisms at scale. Before describing the measurement pipeline and the technical enhancements introduced to support it, we first define the scope of the study and the boundaries under which the resulting measurements should be interpreted.

4.1 Scope

This study examines the use of multiple stateful and stateless tracking mechanisms on highly visited websites. The target population is defined using the Tranco Top-1000 ranking. For each domain, the measurement is restricted to the landing page reached during the automated visit; deeper navigation through the website, authenticated sessions, and interactions with page elements or consent interfaces are outside the scope of the crawl.

The tracking mechanisms considered comprise cookies, web storage, web analytics tools, and active browser fingerprinting. Web storage includes Local Storage, Session Storage, and IndexedDB. Regarding browser fingerprinting, this study focuses exclusively on active techniques observable through browser-side JavaScript instrumentation, specifically Audio Context, Canvas, Canvas Font, WebGL, WebRTC, and JavaScript Enumeration. Consequently, all fingerprinting techniques considered in this work belong to the active category. Passive fingerprinting techniques are outside the scope of the study because they rely on information obtained server-side and cannot be directly observed through the browser-side instrumentation employed in our measurement approach.

Table 2 Scope of previous web tracking measurement studies according to the tracking mechanisms analyzed

Study	Cookies	Storage	Analytics	FP	Canvas	WebRTC	WebGL	Audio	JS Enum.
[17]	✓	–	–	✓	✓	✓	–	✓	–
[7]	–	–	–	✓	–	–	–	–	–
[30]	–	–	–	✓	–	–	–	–	–
[21]	–	–	–	✓	✓	–	–	–	–
[49]	–	✓	✓	✓	✓	✓	✓	–	–
[10]	–	–	✓	✓	✓	–	–	–	–
[2]	–	✓	–	–	–	–	–	–	–
[37]	✓	–	–	–	–	–	–	–	–
[27]	✓	✓	✓	✓	✓	✓	✓	–	–
[46]	–	–	–	✓	–	–	–	–	–
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓

A check mark indicates that the corresponding mechanism was analyzed in the study. FP denotes fingerprinting reported as an overall category, whereas the subsequent columns represent specific active fingerprinting techniques

For web analytics, the study considers Google Analytics, Meta Pixel, Microsoft Universal Event Tracking (UET)/Clarity, and Hotjar. These platforms were selected as the analytics services covered by the detection mechanisms implemented in the analysis pipeline. The study therefore does not attempt to provide exhaustive detection of every existing analytics platform or every possible fingerprinting technique, but rather a joint analysis of the tracking mechanisms explicitly defined above.

The measurements are performed through automated browser crawling. The crawler does not simulate user interactions such as accepting or rejecting consent banners, clicking page elements, scrolling, logging into accounts, or navigating through multiple pages of a website. Similarly, the measurement configuration does not introduce specific mechanisms intended to evade website detection of automated or instrumented browsers. These choices define the conditions under which tracking behavior is observed in this study; their potential impact on detection coverage is discussed separately as a limitation.

4.2 Methodological workflow

Within this scope, the measurement and analysis pipeline consists of five main stages:

1. *Platform enhancement*: prior to the large-scale crawl, we extended OpenWPM's JavaScript instrumentation to collect the information required for the tracking mechanisms considered in this study and enhanced the OpenWPM-Data-Analytics post-processing framework with lightweight detectors for six active fingerprinting categories and four major analytics platforms to support the collection, detection, and analysis of these mechanisms. The resulting measurement and analysis pipeline was validated before its application to the target websites, as detailed in Sect. 5.
2. *Selection of target websites*: we used the Tranco [1] Top-1000 list as the basis for our analysis, as it aggregates and stabilizes rankings from multiple sources to reduce bias [36]. This list has also been used in multiple previous studies, namely Ahmad et al. [2] and Boussaha et al. [10].
3. *Data acquisition*: we executed OpenWPM with the extended JavaScript instrumentation to visit the selected websites and collect the HTTP, cookie, storage, and JavaScript instrumentation data required by the subsequent analysis. The data collection process is described in Sect. 5.3, while the configuration and outcome of the large-scale crawl are reported in Sect. 6.
4. *Data processing*: the raw data collected by OpenWPM was processed using the enhanced OpenWPM-Data-Analytics framework to detect and quantify the tracking mechanisms defined in the scope of the study. Section 5.3 explains this process in detail, while Sect. 6 presents its application to the complete dataset.
5. *Data analysis*: once the collected data had been processed, we analyzed the prevalence and coexistence of the considered tracking mechanisms across the successfully crawled websites. The results of this analysis are presented in Sect. 6.

This methodology enables that our analysis is large-scale, automated, and fine-grained, covering both stateful and stateless tracking techniques. As shown previously in Table 1, the resulting OpenWPM-based measurement and analysis pipeline provides broader coverage of the tracking mechanisms considered in this study than the individual privacy

[1] https://tranco-list.eu.

scanners included in previous works. This methodology provides the basis for the empirical analysis reported in Sect. 6.

5 Improving tracking detection with OpenWPM

In this section, we describe the OpenWPM-based measurement and analysis pipeline used in this study. We first present the relevant features of OpenWPM and the extensions introduced to its JavaScript instrumentation, and then describe the enhancements made to OpenWPM-Data-Analytics for processing and analyzing the tracking mechanisms defined in Sect. 4.1. All source code, implemented improvements, and generated datasets are publicly available in [4].

Figure 1 provides an overview of this technical pipeline, from browser-based data collection with OpenWPM to subsequent processing with the enhanced OpenWPM-Data-Analytics framework. The following subsections describe the corresponding components and their validation in detail.

5.1 Main features of OpenWPM

OpenWPM[2] is an open-source web privacy analysis tool [17] designed to automate the navigation and collection of privacy-related data across multiple websites. The collected data is stored in a database that can subsequently be processed to analyze tracking behaviors.

The tool is written in Python and crawls web pages using Firefox instances controlled through Selenium, either in headful or headless mode. Its architecture is modular, consisting of distributed components that communicate through network sockets. OpenWPM includes several instrumentation mechanisms for data collection, namely HTTP, JavaScript, and cookie instrumentation. Among these, only JavaScript instrumentation is configurable, allowing the set of instrumented browser objects, properties, and functions to be extended according to the requirements of a measurement study.

The default OpenWPM configuration already provides information relevant to several of the tracking mechanisms considered in this study. However, the measurement scope defined in Sect. 4.1 requires additional JavaScript instrumentation and subsequent processing capabilities. We therefore extended the OpenWPM instrumentation configuration and the OpenWPM-Data-Analytics framework as described in the following subsections.

[2] https://github.com/openwpm/OpenWPM.

5.2 OpenWPM enhancements

To support the measurement scope defined in Sect. 4.1, we extended OpenWPM's JavaScript instrumentation configuration. Specifically, the configuration was adapted to instrument the browser objects, properties, and functions required for the subsequent analysis of Canvas, Canvas Font, WebGL, WebRTC, Audio Context, and JavaScript Enumeration fingerprinting, as well as the web storage mechanisms considered in this study. For mechanisms already supported by OpenWPM, such as Canvas, WebGL, and WebRTC, these modifications refine or extend the JavaScript API-access information collected by OpenWPM for our detection criteria rather than introducing support for these mechanisms from scratch. Overall, the extended configuration provides the JavaScript API-access information needed to apply the detection criteria described later in this section.

The resulting instrumentation configuration complements OpenWPM's existing HTTP and cookie instrumentation and provides the browser-side information required by the enhanced OpenWPM-Data-Analytics framework. The instrumentation and post-processing extensions therefore serve different purposes: OpenWPM collects the required browser and network events, whereas OpenWPM-Data-Analytics subsequently processes these events to identify and quantify the tracking mechanisms considered in the study.

The JavaScript instrumentation configuration was also refined by removing Audio Context objects such as `ScriptProcessorNode` and `GainNode`, which are not required by the Audio Context fingerprinting detection criteria employed in this study. The rationale for this configuration is described together with the corresponding detection mechanism below.

As defined in the study scope, these instrumentation extensions do not modify the crawler to evade anti-bot mechanisms or to trigger tracking behaviors that require user interaction, such as those investigated by Annamalai et al. [46]. The resulting system therefore remains an automated scanner-based measurement framework. The implications of these characteristics for measurement coverage are discussed separately as methodological limitations.

5.3 Data acquisition

To automate data collection, we developed a Python script that executes OpenWPM using the instrumentation configuration described above and sequentially visits the websites included in the selected Tranco Top-1000 list. During each visit, OpenWPM collects the browser and network information generated by its enabled instrumentation mechanisms and stores the resulting data in its database for subsequent processing with OpenWPM-Data-Analytics.

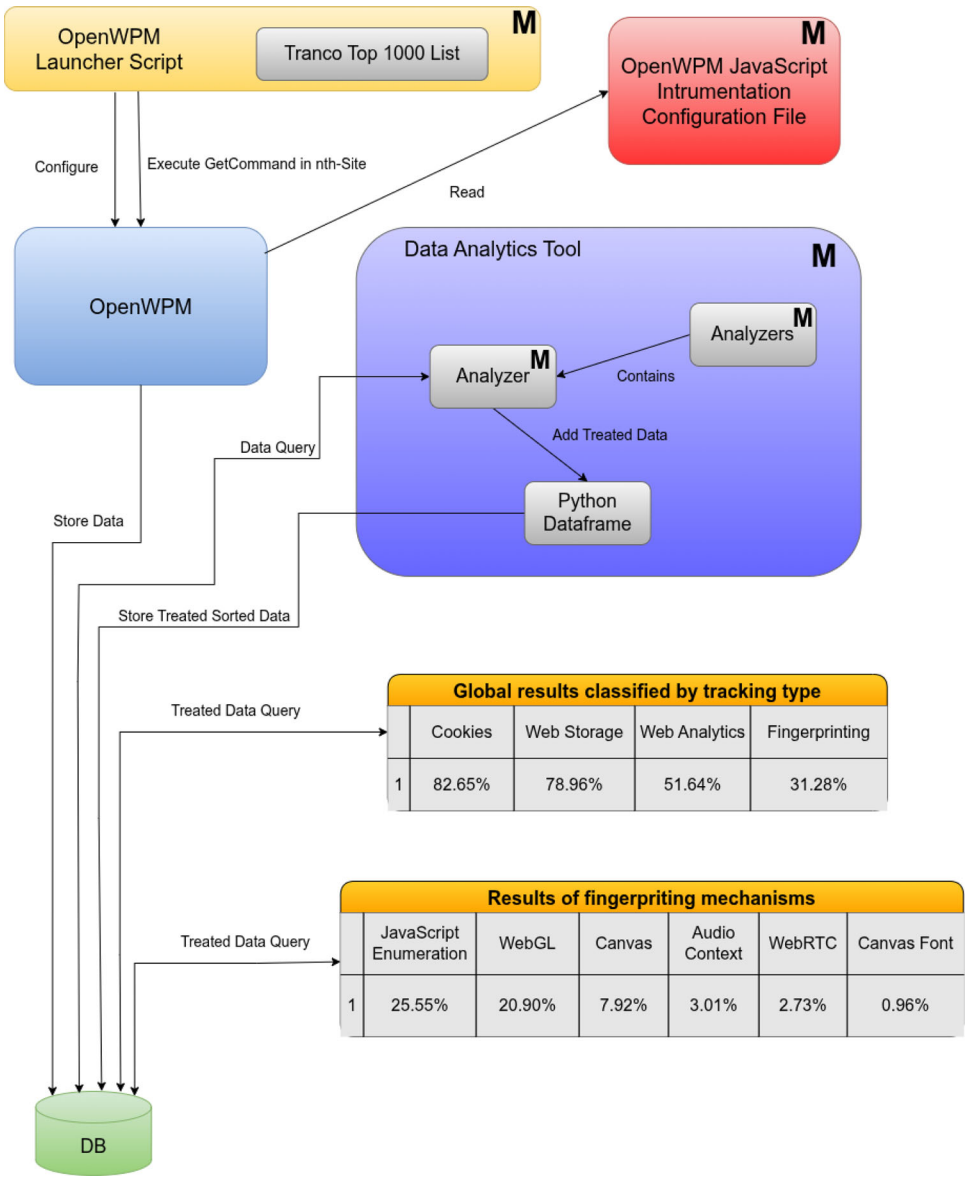


Fig. 1 Overview of the tracking detection process followed in this research. M stands for modified components, namely the OpenWPM launcher script and instrumentation configuration, and the Data Analytics Tool and its analyzers

5.3.1 Ethical considerations

From an ethical standpoint, our crawler-based measurements were designed to minimize disruption to the web ecosystem. Consistent with the study scope defined in Sect. 4.1, the crawling process was intentionally limited to publicly accessible landing pages and their redirects, avoiding deep navigation or intrusive interactions with target websites. This design choice helped reduce the operational impact on servers and follows commonly accepted ethical practices for large-scale web studies [18, 31].

Furthermore, the crawler generated low-rate traffic and avoided repeated or aggressive requests, reducing the likelihood of interfering with normal website operation.

5.4 Data processing

OpenWPM collects raw data from the websites without performing further analysis. We therefore enhanced OpenWPM-Data-Analytics, available in [4], to process the resulting OpenWPM databases and identify and quantify the tracking mechanisms defined in Sect. 4.1. In this way, the data collected through OpenWPM's browser and network instrumentation serves as input to the post-processing stage, where the corresponding detection criteria are applied.

The original OpenWPM-Data-Analytics framework[3] already provided the basic structure required to extract and analyze information from OpenWPM databases and included support for some fingerprinting techniques, such as Canvas

[3] https://github.com/tim-stephenson/OpenWPM-Data-Analytics.

and WebRTC. Its architecture included two different analyzers. The static analyzer identifies tracking-related patterns through string or regular expression matching, whereas the dynamic analyzer maintains a state of matched strings to detect more complex tracking mechanisms, such as chained JavaScript calls commonly used in fingerprinting techniques.

We extended this architecture to support the broader set of tracking mechanisms considered in this study. The main modifications introduced into OpenWPM-Data-Analytics are summarized below, while the following subsections describe the detection criteria applied to each tracking mechanism.

- Two new base analyzer classes were added to process cookies and HTTP requests. These complement the existing framework, whose static analyzer operates on HTTP responses and whose dynamic analyzer queries JavaScript calls. HTTP request analysis is required to identify the domains associated with the web analytics tools considered in this study, while cookie analysis requires querying the corresponding cookie data stored by OpenWPM.
- New analyzer subclasses were implemented and tailored to the specific behavior of the tracking mechanisms considered in this study. Where the original framework already provided analyzers for particular fingerprinting techniques, these were reused and adapted when their original detection criteria required refinement.
- The top-level URL of the visited website was incorporated into the composite key of the SQL results table. This facilitates the association of detected tracking activity with the corresponding crawled website and enables first- and third-party contexts to be distinguished during subsequent analysis.
- The analysis framework was extended to quantify cookie and web storage activity. Whereas the original implementation primarily recorded whether a given tracking mechanism was detected on a website, the enhanced version also supports counting the corresponding cookies and web storage accesses.

5.4.1 Web analytics tools

The presence of the web analytics platforms considered in the study was detected by analyzing the domains contacted through HTTP requests collected by OpenWPM.

For this purpose, the domain of each HTTP request was matched against 11 regular expressions covering known domains associated with the selected analytics services. Both exact and partial matches were considered to account for subdomains and service-specific request patterns. The domains and request patterns were identified using *WhoTracksMe*[4], the official documentation of each platform, and manual inspection of the source code of websites using these services.

The following detection criteria were applied to the four web analytics platforms included in the study:

- *Google Analytics* [25]. Detection considers requests associated with seven domains: *googleanalytics.com*, *google-analytics.com*, *googletagmanager.com*, *googletagservices.com*, *doubleclick.net*, *invitemedia.com*, and *2mdn.net*.
- *Meta Pixel* [41]. Detection considers requests to *facebook.com* and *facebook.net* matching the Meta Pixel request patterns */tr?id=identifier&ev=event* and */tr/?id=identifier&ev=event*. The *event* component corresponds to events defined in the official Meta Pixel documentation [41], whereas the *identifier* represents the customer-specific identifier.
- *Microsoft Universal Event Tracking (UET)* [42] and *Clarity* [43]. Detection considers requests associated with *bing.net* and *clarity.ms*.
- *Hotjar* [29]. Detection considers requests associated with *static.hotjar.com*.

5.4.2 Web fingerprinting

The detection of browser fingerprinting involves several steps and is based on the JavaScript API accesses collected through the OpenWPM instrumentation described above. First, the analyzer checks the instantiation of certain classes, sometimes with specific parameters. Next, it verifies whether certain functions modify the state of the corresponding objects. Finally, the detector analyzes the extractor functions to determine whether they obtain unique characteristics or hashes that can be used for identification. However, in the case of enumeration-based fingerprinting, verifying the presence of class instantiations or state-modifying functions may not be necessary.

The detection criteria applied to each fingerprinting technique considered in this study are described below. The criteria were designed to balance the detection of fingerprinting behavior with the minimization of false positives. Their initial description is based on the information provided by the Mozilla Developer Network (MDN) Web Docs [45], DuckDuckGo Tracker Radar[5], and Browserize [60]. Depending on the technique, detection relies on combinations of object instantiation, state-modifying operations, information-access functions, extraction functions, or

[4] https://whotracks.me/trackers.html.

[5] https://github.com/duckduckgo/tracker-radar-collector.

Table 3 Websites with known trackers used for the validation of the tool, indicating the tracking mechanisms expected on each website.

Website	Fingerprinting							Web storage			Web analytics tools				Cookies	
	Audio context	Canvas	Canvas font	WebGL	WebRTC	JS enumeration		Local	Session	IndexedDB	Google analytics	Meta Pixel	Microsoft UET	Hotjar	First-Party	Third-Party
IAmUnique	✓	✓	✓	✓	×	✓		✓	✓	✓	×	×	×	×	✓	×
BrowserLeaks	✓	✓	×	✓	✓	✓		×	×	×	×	×	×	×	×	×
Browserize	×	✓	✓	✓	✓	✓		✓	✓	✓	×	×	×	×	×	×
CoverYourTracks	×	×	×	×	×	×		×	×	×	×	×	×	×	✓	✓
Browserscan	×	✓	×	✓	×	✓		✓	×	×	✓	×	✓	×	✓	✓
CurseForge	×	×	×	×	×	✓		✓	✓	×	✓	✓	✓	×	✓	✓
Imgur	×	×	×	×	×	×		✓	✓	×	✓	✓	×	×	✓	✓
NexusMods	×	×	×	✓	✓	×		✓	✓	✓	✓	×	×	✓	✓	✓

enumeration behavior. The specific criteria applied to each fingerprinting category are as follows:

- *Audio Context*. Audio Context fingerprinting is detected by analyzing operations involving the *AudioContext* or *OfflineAudioContext* APIs. The detector first checks whether at least one relevant audio-processing object is created. Specifically, the detection considers methods such as *createOscillator*, *createDynamicCompressor*, *createAnalyser*, *audioWorklet*, *createGain*, and *createScriptProcessor*. Then, it checks whether an audio extraction function is invoked, such as *AudioBuffer.getChannelData*, *AudioBuffer.copyFromChannel*, *AnalyserNode.getFloatFrequencyData*, or *AnalyserNode.getByteFrequencyData*. Alternatively, the detection also considers enumeration-related API accesses, including *destination* and *listener* from *AudioContext* or *OfflineAudioContext*.
- *Canvas*. Canvas fingerprinting is detected by first checking whether the canvas element is hidden or whether its height or width is smaller than 16 pixels. The detector then verifies whether more than two different colors are drawn, whether a text string of at least 10 characters is rendered, or whether a rectangle is drawn. Finally, the detection requires the use of an extractor function, specifically the *toDataURL* and *toBlob* methods from *HTMLCanvasElement*, or the *getImageData* method from *CanvasRenderingContext2D*.
- *Canvas Font*. Canvas Font fingerprinting is detected by analyzing calls to `measureText`. The detector checks whether at least 50 different fonts are set and whether at least one text element has been measured more than 50 times. These thresholds were selected to distinguish systematic font enumeration from ordinary font measurements and were subsequently refined through the analysis of JavaScript execution traces and manual inspection in Firefox.
- *WebGL*. WebGL fingerprinting is detected by analyzing calls to *WebGLRenderingContext* and *WebGL2RenderingContext*. The detector considers *getParameter* with the arguments 37445 and 37446, requesting debug information; *getSupportedExtensions* and *getExtension*, to enumerate the available extensions; *getContextAttributes*, to obtain default attributes supported by the context; or *readPixels*, an extraction function. Additionally, the analyzer checks whether a WebGL context is created through *HTMLCanvasElement.getContext* using the *'webgl'* or *'webgl2'* arguments. The detection also verifies calls to the *toDataURL* or *toBlob* methods from the *HTMLCanvasElement* class.
- *WebRTC*. WebRTC fingerprinting is detected by analyzing operations involving *RTCPeerConnection*. Specifi-

cally, the detector checks for *createDataChannel*, used to create a data channel; *createOffer*, which generates an SDP offer; and *onicecandidate*, used to manage new ICE candidates.

– *JavaScript Enumeration*. JavaScript enumeration is detected when more than 12.5% of the variables, properties, and functions containing user-related information are accessed. In total, 90 objects are monitored, mainly including the full hierarchy of the *window.navigator* variable, the *window.Intl* function, and the *window.performance.timing* variable, as well as *window.history* and *window.screen.pixelDepth*. The 12.5% threshold was selected to limit false positives while retaining sensitivity to systematic enumeration behavior.

The initial criteria and heuristic thresholds described above were subsequently refined by analyzing JavaScript execution traces collected by OpenWPM and manually inspecting the corresponding browser behavior in Firefox.

5.4.3 Web storage

Web storage usage is detected by analyzing accesses to the corresponding browser-side global objects collected through OpenWPM's JavaScript instrumentation. Specifically, the analysis considers *window.localStorage* for local storage, *window.sessionStorage* for session storage, and *window.indexedDB* for the IndexedDB global object. For each visited website, the enhanced OpenWPM-Data-Analytics framework identifies the storage mechanisms accessed and records the corresponding activity for subsequent quantification.

5.4.4 Cookies

Cookies are first classified according to their relationship with the visited website. To distinguish between first- and third-party cookies, the domain associated with each cookie is compared with the top-level domain of the visited website. Cookies associated with the same domain are classified as first-party cookies, whereas cookies associated with a different domain are classified as third-party cookies. Cookies are further characterized according to selected attributes recorded by OpenWPM. A cookie is classified as *Host-only* when the corresponding host-only attribute is set, as *Session* when it has a session lifetime, and *Secure* when either the *Secure* or *HttpOnly* attribute is present. Combining the first- and third-party classification with these attribute-based categories results in the six cookie categories considered in the subsequent analysis.

Finally, the heuristic detectors incorporated into OpenWPM-Data-Analytics automate the processing of OpenWPM crawling datasets according to the detection criteria described above. Their modular design also facilitates the adaptation or extension of these criteria as browser APIs and tracking implementations evolve.

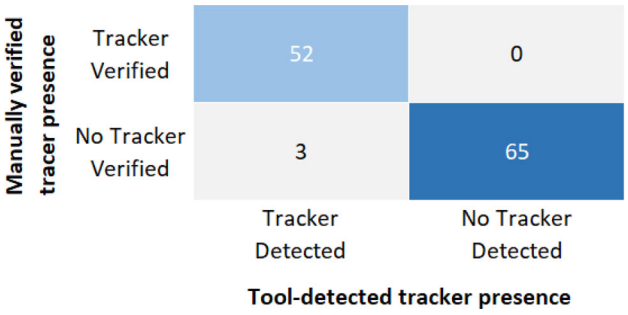


Fig. 2 Resulting confusion matrix from the validation with known tracker sites

5.5 Validation of the measurement and analysis pipeline

After defining and refining the detection criteria described above, we evaluated the resulting measurement and analysis pipeline. The validation primarily focused on the analyzers implemented or adapted in OpenWPM-Data-Analytics. In addition, manual inspection of the data collected during the crawls was used to verify that the extensions introduced into OpenWPM's JavaScript instrumentation captured the information required by these analyzers. The validation therefore considered both the data collection and post-processing components introduced or modified in this work.

The validation strategy used two complementary sets of websites: one containing manually verified tracking mechanisms and another containing no observable instances of the tracking mechanisms considered in the study. The first set was used to evaluate both successful detections and missed tracking instances, whereas the second provided a larger negative sample for assessing false positives.

5.5.1 Validation with websites containing known tracking mechanisms

Table 3 presents the websites selected for the validation together with the tracking mechanisms manually verified before executing the automated analysis. Manual inspection was performed using Firefox with its security protections disabled and included JavaScript code, network requests, cookies, and browser storage. The websites were intentionally selected to cover different combinations of the mechanisms considered in the study, including dedicated fingerprinting demonstration websites and real-world websites exhibiting analytics, cookie, and storage activity.

Table 4 Metrics regarding the validity of the sample containing known trackers

Metric	Value	Confidence interval (95%)
False Positive Rate (FPR)	0.00%	[0.00%, 5.52%]
False Negative Rate (FNR)	5.45%	[1.14%, 14.90%]
Sensitivity	94.55%	[84.70%, 98.90%]
Specificity	100.00%	[94.48%, 100.00%]
Precision	100.00%	[93.15%, 100.00%]
Accuracy	97.50%	[92.88%, 99.40%]

For fingerprinting, the inspection considered the JavaScript method calls and parameters associated with the detection criteria described above. Web analytics platforms were verified through the corresponding network requests and domains, while cookies and web storage were inspected directly through the browser and network information. When JavaScript code was obfuscated, the relevant methods and arguments were manually traced to verify the observed behavior.

Figure 2 shows the resulting confusion matrix. Each observation corresponds to one tracking-mechanism category evaluated for one website, resulting in 120 evaluated instances. Of these, 52 were true positives, 65 were true negatives, and 3 were false negatives; no false positives were observed. The three false negatives correspond to Canvas fingerprinting on IAmUnique, BrowserLeaks, and Browserize.

These false negatives result from the Canvas detection heuristic described above, which requires the canvas elements to be hidden or sufficiently small. On the affected fingerprinting demonstration websites, the relevant canvas elements were intentionally visible and rendered at standard dimensions. In contrast, the canvas elements used by Browserscan satisfied the size condition and were correctly detected.

These results reveal a specific limitation of the adopted Canvas heuristic: fingerprint extraction performed from visible, standard-sized canvas elements may remain undetected. This limitation is discussed further in the construct-validity analysis below.

From the confusion matrix, several statistical metrics were computed, as shown in Table 4. Confidence intervals were estimated using the Clopper-Pearson exact method [12], as some observed proportions were located near the boundaries of the parameter space (e.g., False Positive Rate (FPR) = 0 and Precision = 1). In such cases, exact binomial intervals provide guaranteed coverage and avoid the inaccuracies associated with normal approximations such as the Wald interval. Although they are more conservative than score-based methods such as Wilson intervals, they are generally preferred when dealing with extreme proportions or small sample sizes.

Concretely, the confidence intervals were computed based on the binomial formulation of each metric, where the numerator can be interpreted as the number of successes (x) and the denominator as the number of trials (n). For example, for the FPR, x corresponds to the number of false positives, whereas n corresponds to the sum of false positives and true negatives.

For this validation set, the resulting detector achieved a sensitivity of 94.55%, a specificity and precision of 100%, and an overall accuracy of 97.50%.

5.5.2 Validation with websites without trackers

A second validation was conducted using a dataset of 50 websites for which no instances of the tracking mechanisms considered in this study were identified during manual inspection. The selected websites mainly corresponded to open-source and privacy-oriented projects, including Qubes OS, Mullvad, and GNU-related web pages. As in the previous validation, the manual inspection was performed before the automated analysis. No fingerprinting mechanisms, analytics tools, cookies, or storage-based tracking elements were identified during this inspection.

Figure 3 presents the confusion matrix resulting from this validation. All 750 evaluated tracking-mechanism instances were classified as true negatives, and no false positives were detected.

The resulting statistical metrics are shown in Table 5. Since no positive instances were present in the dataset, the False Negative Rate (FNR), sensitivity, and precision metrics are mathematically undefined.

For this validation set, the analysis yielded an FPR of 0%, a specificity of 100%, and an overall accuracy of 100%, providing additional evidence of the detector's ability to avoid false positives for the tracking mechanisms considered.

To further assess the consistency of the validation results, both validations using websites with and without trackers were repeated using a sample corresponding to 10% of the websites in each set. The repeated analyses produced no false positives and yielded results consistent with those obtained in the original validation runs.

Overall, the validation results indicate that the enhanced OpenWPM-Data-Analytics framework provides highly reli-

Table 5 Metrics regarding the validity of the sample without trackers

Metric	Value	Confidence interval (95%)
False Positive Rate (FPR)	0.00%	[0.00%, 0.49%]
Specificity	100.00%	[99.51%, 100.00%]
Accuracy	100.00%	[99.51%, 100.00%]

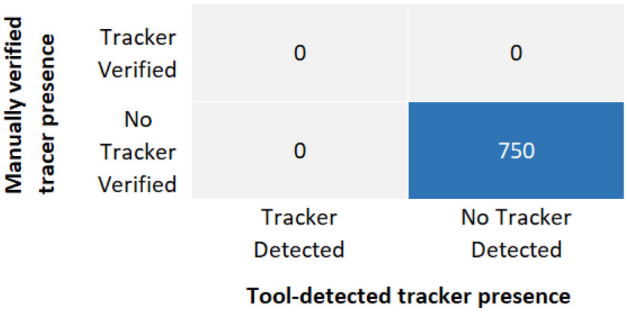


Fig. 3 Resulting confusion matrix from the validation using websites with no identified tracking mechanisms

able detection for the tracking mechanisms considered in the validation sets, particularly in terms of avoiding false positive. The observed false negatives were limited to specific Canvas fingerprinting scenarios in which visible, standard-sized canvas elements did not satisfy the assumptions of the adopted detection heuristics.

5.5.3 Construct validity and limitations

Despite the favorable validation results, several limitations related to construct validity and measurement coverage must be considered when interpreting the results.

Regarding fingerprinting detection, the main limitation observed during validation concerns the Canvas heuristic. The current detection criteria assume that fingerprinting-related canvas elements are hidden or rendered with very small dimensions. Consequently, fingerprint extraction performed using visible, standard-sized canvas elements may remain undetected, as observed for the three Canvas false negatives in the validation set.

Several fingerprinting detectors also rely on heuristic thresholds intended to reduce false positives. For example, Canvas Font detection requires the use of at least 50 different fonts, while JavaScript Enumeration requires access to more than 12.5% of the monitored objects, properties, and functions. These values were defined empirically for this study rather than adopted from thresholds established in previous literature. Fingerprinting implementations that operate below these thresholds may therefore remain undetected. Lower thresholds could increase sensitivity but would also increase the risk of classifying legitimate browser API usage as fingerprinting.

Although the heuristic thresholds were empirically selected rather than systematically optimized, the validation results suggest that these thresholds provide a reasonable trade-off between detection sensitivity and false positive avoidance for the evaluated datasets.

Web analytics detection depends on known domains and request patterns associated with the four analytics platforms considered in the study. Although these patterns were identified using multiple sources, undocumented domains, deployment-specific endpoints, or future changes in the services may remain undetected. The reported prevalence of these analytics platforms should therefore be interpreted with respect to the detection rules available at the time of the measurement.

For cookies and browser storage mechanisms, the current implementation only verifies their presence and does not distinguish between functional and tracking-oriented usage. Future versions of the tool are planned to incorporate identifier and string analysis techniques capable of identifying suspicious tracking-related values stored in these elements.

Finally, the automated crawling approach inherits limitations associated with scanner-based measurements. As defined in Section 4.1, the crawler does not perform user interactions such as accepting consent banners, logging into accounts, scrolling, or navigating through multiple pages, nor does it incorporate mechanisms specifically intended to evade anti-bot detection. Tracking behaviors that are activated only under these conditions may therefore remain unobserved. This limitation is particularly relevant for fingerprinting, since previous work has shown that interaction-dependent fingerprinting behavior may be missed by automated privacy scanners [46].

5.6 Performance of the tools

We also evaluated the computational overhead introduced by the proposed extensions and the suitability of the resulting measurement and analysis pipeline for conducting large-scale studies. The following subsections report the performance of the modified OpenWPM instrumentation and the enhanced OpenWPM-Data-Analytics framework.

The experiments were conducted on a system equipped with an Intel i7-13700H CPU, 16 GiB of LPDDR5 RAM, 12 GiB of swap space, and 1 TB of NVMe storage.

5.6.1 OpenWPM

To evaluate the performance impact of the new instrumentation introduced in this paper, we compared the default JavaScript instrumentation configuration of OpenWPM with the modified one over the top 100 websites.

With the default configuration, the crawl required 31 minutes and 11 seconds, with a peak memory usage of 1556 MiB and a database size of 463 MiB. In contrast, the execution with the modified configuration required 31 minutes and 58 seconds, with a maximum memory usage of 1872 MiB and a database size of 500 MiB. In both cases, the average crawling time per website was approximately 15 seconds, including a sleep interval of 3 seconds between crawls.

The comparison shows that the modified instrumentation introduces only a limited increase in crawling time and database size for the evaluated dataset, while memory consumption increases by approximately 20%. Therefore, the main performance cost of the additional instrumentation is memory usage, which may reduce the number of browser instances that can be executed in parallel under constrained system resources. Overall, the observed overhead remains compatible with the type of large-scale OpenWPM-based measurements considered in this study.

5.6.2 OpenWPM-data-analytics

Unlike the OpenWPM performance evaluation, this analysis is not intended to compare the original and enhanced versions of OpenWPM-Data-Analytics. Instead, its purpose is to characterize the computational cost of the post-processing stage within the complete measurement pipeline used in this study. Since the enhanced OpenWPM-Data-Analytics framework processes the database generated by the extended instrumentation, evaluating it on data collected with the original instrumentation would not provide an equivalent baseline for several of the tracking mechanisms considered in our analysis.

The performance of the enhanced OpenWPM-Data-Analytics framework was evaluated on the same system using the database generated by the modified OpenWPM configuration for the previous crawl of 100 websites. Processing this database required 2 minutes and 35 seconds, increased its size by 6 MiB, and reached a maximum memory consumption of 454 MiB.

These measurements provide a performance baseline and show that the post-processing stage can be integrated into the OpenWPM-based pipeline under the evaluated configuration and dataset. The current implementation does not support parallel processing, which limits its ability to exploit multi-core systems directly. Large datasets can nevertheless be processed in separate batches, for example by partitioning websites according to ranking intervals or other dataset-level criteria. Parallel processing could be incorporated in future versions to improve scalability further.

Additional experiments showed that processing cost depends not only on the number of websites but also on the amount of tracking-related activity contained in the collected data. For example, processing the validation dataset comprising 8 websites with numerous known tracking mechanisms required 12 minutes and 13 seconds. The post-processing results increased the size of the 139.3 MiB input database by 2%.

Memory requirements also depend on the size of the intermediate query results, particularly for queries involving the *javascript* table, which is typically the largest table in the OpenWPM database. For instance, processing the 2.8 GiB database generated during the large-scale measurement described in Sect. 6 required up to 1 GiB of memory.

6 Web tracking analysis on top websites

After validating the measurement and analysis pipeline described in the previous section, we applied it to the Tranco Top-1000 websites. The modified OpenWPM configuration was used to collect the corresponding browsing data, which was subsequently processed using the enhanced OpenWPM-Data-Analytics framework. This section presents the resulting dataset and analyzes the prevalence and distribution of the tracking mechanisms considered in the study.

The complete crawl required 7 hours and 6 minutes and generated an OpenWPM database of 2883 MiB. Processing the collected data with OpenWPM-Data-Analytics required 31 minutes and 43 seconds, reached a maximum memory usage of 1024 MiB, and increased the database size by 40 MiB. The Tranco list used in this study is publicly available under the identifier 7XX3X[6].

Of the 1,000 domains included in the Tranco list, 732 were successfully crawled, yielding a final sample size of 732 websites for the analysis presented below. Figure 4 shows the distribution of successful and unsuccessful crawls across rank intervals. Most failures resulted from domains that could not be resolved to valid IP addresses, while a smaller number were associated with unavailable websites or JavaScript execution errors.

A substantial proportion of the unsuccessful entries corresponded to infrastructure-oriented services (CDN, DNS, third-party services that cannot be accessed directly) rather than conventional user-facing web pages. Examples include Azure Front Door (*a-msedge.net*), Akamai CDN servers (*akamai.net* and *akamaiedge.net*), Windows Update (*windowsupdate.net*), and DNS root servers (*root-servers.net*). Their exclusion therefore does not necessarily correspond to

[6] https://tranco-list.eu/list/7XX3X/full.

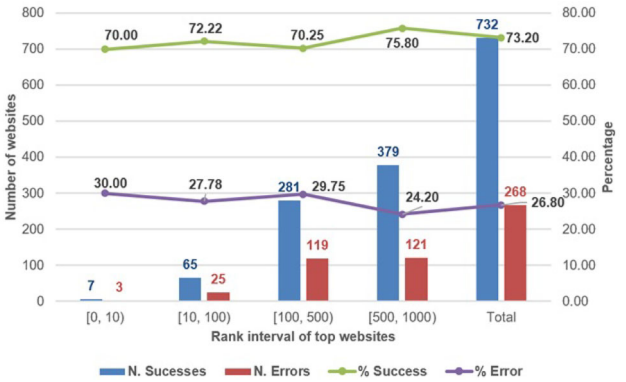


Fig. 4 Number and percentage of websites successfully and unsuccessfully crawled, broken down by the considered rank intervals from the original Tranco list

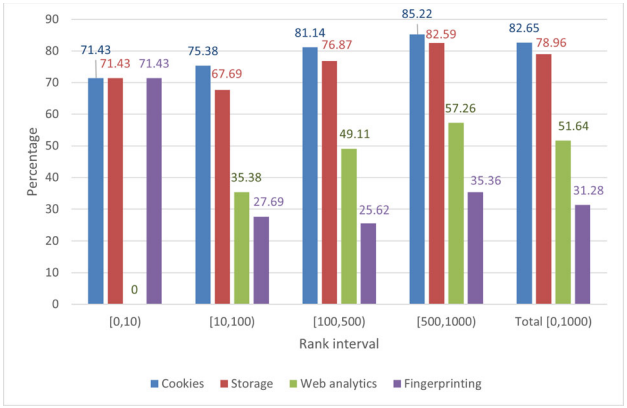


Fig. 5 Prevalence of each kind of tracker for each subset of websites

the loss of accessible websites on which the tracking mechanisms considered in this study could have been measured.

The final dataset includes 732 successfully crawled websites, corresponding to 73.2% of the original Tranco Top-1000. Although the 26.8% crawl failure rate may introduce sampling bias, the predominance of infrastructure-oriented services among the unsuccessful entries may reduce its effect on the measurement of tracking prevalence. In addition, the observed results remained broadly consistent across the analyzed popularity intervals. For a sample of 732 websites, the maximum margin of error at a 95% confidence level is approximately ±3.62%.

For clarity, throughout this section, higher-ranked websites refer to more popular websites with lower numerical Tranco rank values, whereas lower-ranked websites refer to less popular websites with higher numerical rank values.

Figure 5 presents the global results for the four main categories considered in the study: cookies, web storage, web analytics, and browser fingerprinting. The results show substantial differences in prevalence across these mechanisms.

As shown in Fig. 5, cookies are the most prevalent mechanism, being observed on 82.65% of the successfully crawled websites. Web storage follows closely, with a prevalence of 78.96%. Web analytics tools are detected on 51.64% of websites, while active browser fingerprinting is observed on 31.28%. The following subsections examine each category in greater detail.

Table 6 reports the corresponding confidence intervals. The intervals are considerably wider for the highest-ranked subsets because of their smaller numbers of successfully crawled websites, whereas the larger rank intervals provide narrower estimates. For the complete sample, the confidence intervals are [79.71%, 85.33%] for cookies, [75.83%, 81.86%] for web storage, [47.95%, 55.31%] for web analytics, and [27.94%, 34.78%] for active browser fingerprinting.

6.1 Cookies

Figure 5 shows that cookies are highly prevalent across all ranking intervals, being observed on 82.65% of the successfully crawled websites. Across the analyzed ranking intervals, their prevalence increases toward lower-ranked websites, from 71.43% in [0,10) to 85.22% in [500,1000).

Figure 6 provides a more detailed analysis according to cookie origin. First-party cookies were observed on 81.28% of the analyzed websites, with prevalence ranging from 71.43% in [0,10) to 84.43% in [500,1000).

Third-party cookies were detected on 57.38% of the websites. Their prevalence increases toward lower-ranked websites, from approximately 28% in the [0,10) interval to approximately 68% in the [500,1000) interval.

Figure 7 further characterizes cookies according to the attributes considered in the analysis. Cookies classified as Secure were observed on 66.39% of websites in first-party contexts and 53.69% in third-party contexts. These values also indicate that a substantial proportion of websites do not exhibit cookies classified as Secure, particularly in third-party contexts. Host-only cookies were observed on 54.10% and 33.61% of websites in first- and third-party contexts, respectively. Session cookies showed the largest difference between the two origins, being observed on 55.05% of websites in first-party contexts and 27.19% in third-party contexts.

Regarding rank intervals, the prevalence of first-party cookies remains comparatively stable, with the interval [10,100) showing the lowest values for the analyzed categories. Third-party cookies display greater variation, with substantially lower prevalence among the highest-ranked websites than in the remaining intervals. Among the attribute-based categories, Secure third-party cookies show the largest difference between the highest- and lower-ranked intervals.

Table 6 Confidence intervals (95%) of each kind of tracker for each subset of websites

Rank interval	Fingerprinting	Web analytics	Cookies	Storage
[0, 10)	[29.04, 96.33]	[0.00, 40.96]	[29.04, 96.33]	[29.04, 96.33]
[10, 100)	[17.31, 40.19]	[23.92, 48.23]	[63.13, 85.23]	[54.95, 78.77]
[100, 500)	[20.62, 31.15]	[43.12, 55.12]	[76.07, 85.54]	[71.49, 81.67]
[500, 1000)	[30.54, 40.40]	[52.10, 62.29]	[81.25, 88.64]	[78.39, 86.27]
Total [0, 1000)	[27.94, 34.78]	[47.95, 55.31]	[79.71, 85.33]	[75.83, 81.86]

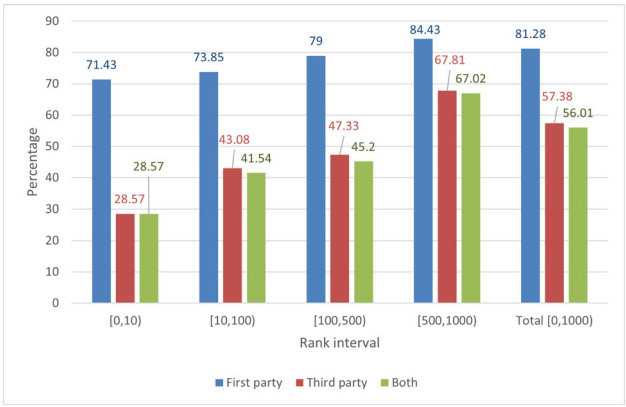


Fig. 6 Percentage usage of the cookies classified by the origin server for each subset of web pages

6.2 Web storage

Figure 5 shows that web storage is also highly prevalent across the analyzed websites, being observed on 78.96% of the successfully crawled websites. Its prevalence ranges from 71.43% in [0,10) to 8 2.59% in [500,1000).

Figure 8 provides a more detailed breakdown of the storage mechanisms considered in the study. Local Storage was observed on 76.50% of the analyzed websites, while Session Storage was observed on 70.36%. Regarding rank intervals, both technologies show their lowest and their highest prevalence in intervals [10,100) and [500,1000), respectively.

IndexedDB was considerably less prevalent, being observed on 27.60% of the analyzed websites. Its prevalence also increases toward lower-ranked websites.

6.3 Web analytics tools

As shown in Fig. 5, web analytics platforms were detected on 51.64% of successfully crawled websites. Their prevalence varies substantially across ranking intervals, ranging from 0% in [0,10) to 57.26% in the lower-ranked intervals.

Figure 9 provides a breakdown by analytics platform. Google Analytics stands out as the most prevalent web analytics tool, being detected on 50.82% of the analyzed websites and reaching 56.46% in the interval [500,1000). The remaining platforms are considerably less prevalent: Meta Pixel was detected on 8.88% of websites, reaching 11.08% in [500,1000); Hotjar on 3.69%, with a maximum of 4.22% in [500,1000); and Microsoft UET on 3.01%, also reaching its highest prevalence in [500,1000).

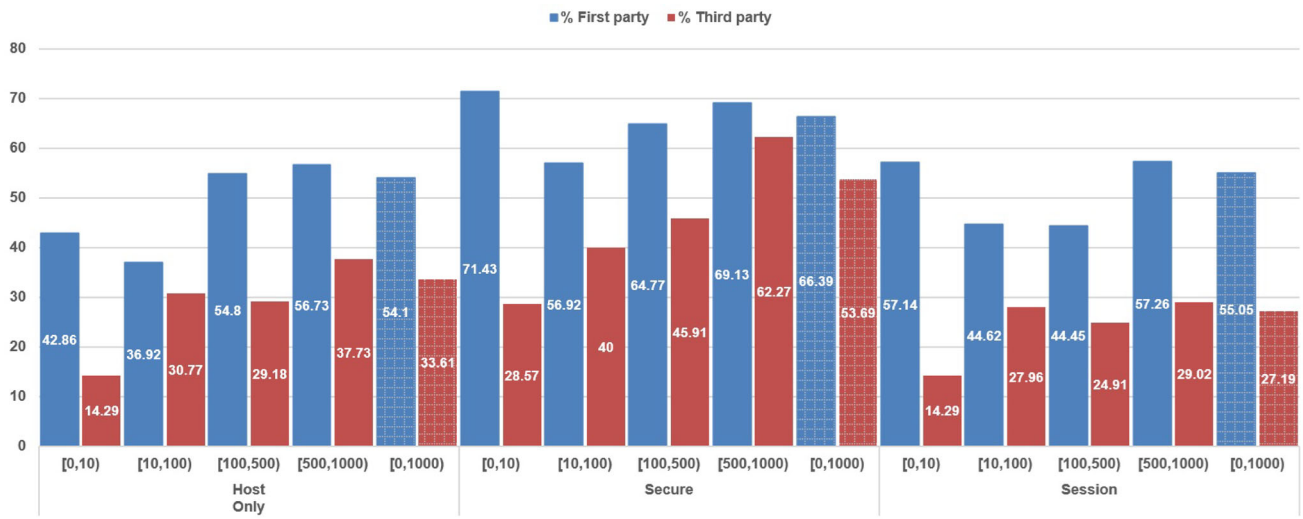


Fig. 7 "Percentage of successfully crawled websites exhibiting host-only, Secure, and Session cookies, classified by the origin, for each ranking interval

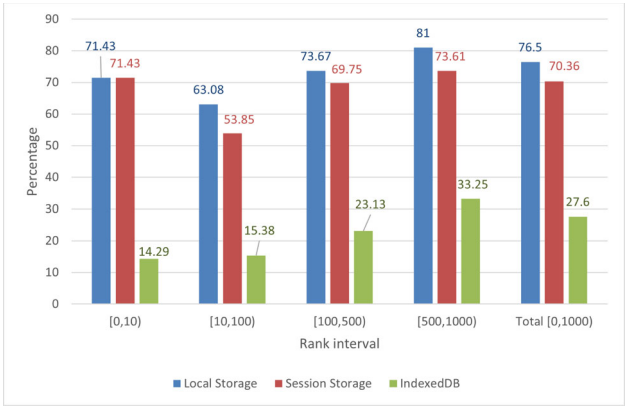


Fig. 8 Percentage usage of web storage access for each subset of websites

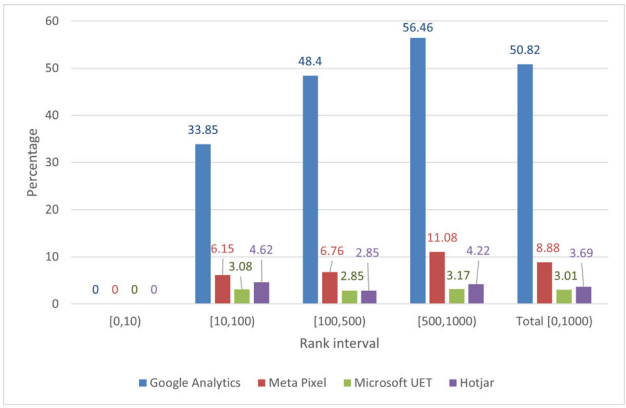


Fig. 9 Percentage usage of each web analytics tool considered for each subset of websites

6.4 Web fingerprinting

As shown in Fig. 5, active browser fingerprinting is the least prevalent of the four main mechanisms considered in the study, being detected on 31.28% of the successfully crawled websites. Fingerprinting is observed across all ranking intervals, although its prevalence varies substantially. In particular, the [0,10) interval shows the highest prevalence at 71.43%, whereas the remaining intervals range from 25.62 to 35.36%.

Figure 10provides a more detailed breakdown of the active fingerprinting techniques detected. JS Enumeration (25.55%) and WebGL (20.90%) are the most prevalent techniques on the analyzed websites. In contrast, Audio Context (3.01%), WebRTC (2.73%), and particularly Canvas Font fingerprinting (0.96%) are considerably less prevalent.

The distribution of individual fingerprinting techniques also varies across ranking intervals. Among the successfully crawled websites in the [0,10) interval, only three of the six considered techniques were detected: JS Enumeration, WebGL, and Canvas. The [100, 500) interval is the first

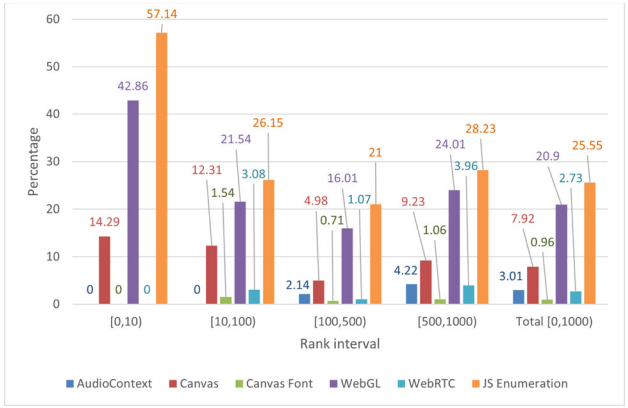


Fig. 10 Prevalence of each type of fingerprinting technique for each subset of websites

range in which all six fingerprinting techniques are observed. Thus, the lower-ranked intervals exhibit a broader variety of the fingerprinting techniques considered in the study.

6.5 Coexistence of tracking mechanisms

As shown in Fig. 11, the simultaneous presence of multiple tracking mechanisms is common among the analyzed websites. Specifically, 45.36% of websites exhibit exactly three of the four mechanism categories considered in the study, while an additional 16.12% exhibit all four tracking-mechanism categories. In contrast, 13.11% of the analyzed websites show none of the considered mechanisms, and 6.83% exhibit only one. Overall, 61.48% of the websites exhibit at least three of the four tracking-mechanism categories considered in the study.

The distribution also varies across ranking intervals. Among the successfully crawled websites in [0,10), none exhibit all four tracking mechanisms, while 57.14% exhibit exactly three. In [10,100), 18.46% show none of the considered mechanisms, whereas 7.69% exhibit all four. In the subsequent intervals, the proportion of websites exhibiting exactly three or all four mechanisms increases, reaching 48.28% and 19.79%, respectively, in [500,1000).

When grouping the websites exhibiting three and four tracking mechanisms, the proportion of websites exhibiting at least three mechanism categories is 57.14% in [0,10), 47.69% in [10,100), 55.87% in [100,500), and 68.07% in [500,1000). Across the complete dataset, this proportion is 61.48%.

To characterize this coexistence in greater detail, Fig. 12 reports the specific combinations of the four mechanism categories. The most frequent combination, observed on 30.87% of websites, consists of web analytics, cookies, and web storage. All four categories, including active browser fingerprinting, coexist on 16.12% of websites. Cookies and web storage without analytics or fingerprinting account for

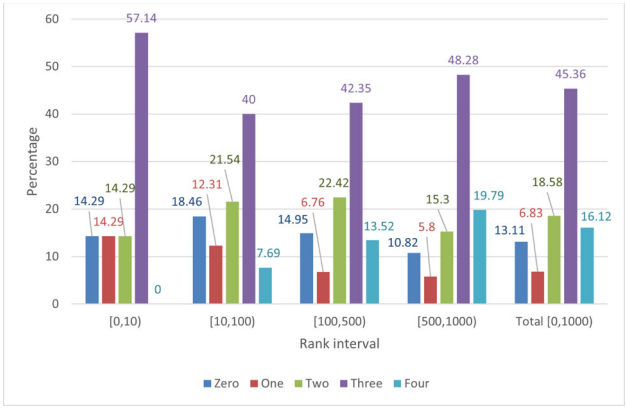


Fig. 11 Percentage usage of the number of different tracking mechanisms for each subset of websites

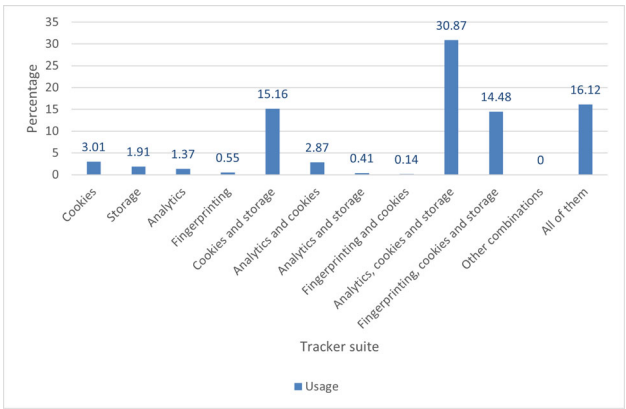


Fig. 12 Prevalence of the different combinations of the four tracking mechanism categories considered

15.16%, while the combination of fingerprinting, cookies, and web storage accounts for 14.48%. Individual categories are considerably less frequent: cookies alone occur on 3.01% of websites, web storage alone on 1.91%, web analytics alone on 1.37%, and fingerprinting alone on 0.55%.

Beyond the descriptive prevalence analysis, we examined whether the number of detected tracking mechanisms is associated with website popularity rank.

6.6 Correlation analysis

Finally, we analyzed the relationship between website popularity rank and the number of detected tracking mechanisms using both Pearson and Spearman correlation coefficients. Pearson correlation was used to assess linear relationships, whereas Spearman correlation was used to assess monotonic relationships between the variables.

For the Pearson correlation, the website rank of each successfully crawled website was paired with the corresponding number of detected tracking mechanisms for the analyzed category. For the Spearman correlation, an average website

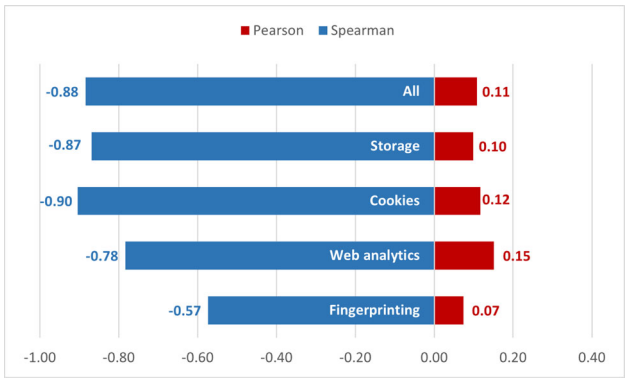


Fig. 13 Pearson and Spearman correlation coefficients between the site rank and the number of tracking mechanisms

rank was computed for each observed number of tracking mechanisms, and the correlation was then calculated using these average ranks (Fig. 13).

The Pearson coefficients indicate only weak linear relationships between website rank and the number of detected tracking instances. The Spearman coefficients show stronger monotonic relationships for some mechanisms, indicating that larger numbers of detected instances tend to occur more frequently among lower-ranked sites. Fingerprinting shows a different pattern, influenced by its comparatively high prevalence among some of the highest-ranked websites, as also observed in Fig. 5.

7 Discussion

Our results provide an updated view of how multiple stateful and stateless tracking mechanisms coexist across highly visited websites. Cookies and web storage remain the most prevalent mechanisms, while web analytics platforms are detected on approximately half of the analyzed websites and active browser fingerprinting on nearly one third. More importantly, the joint analysis shows that these mechanisms frequently coexist on the same websites rather than appearing in isolation.

As discussed in Section 3, previous measurement studies , fingerprinting [10, 21, 30, 46], or storage APIs [2]. Only a limited number of works jointly analyze several tracking vectors [17, 27, 49]. By studying these mechanisms together, our work provides a broader perspective on how modern websites maintain persistent identification and behavioral profiling capabilities through the combined use of multiple tracking technologies. This trend is consistent with recent studies describing an evolution toward more diversified tracking strategies [27, 46, 49]differ substantially in the tracking mechanisms they examine and in the granularity with which they report their results. The present study

complements this literature by measuring cookies, web storage, web analytics, and several active browser fingerprinting techniques within the same OpenWPM-based measurement setting, allowing both their individual prevalence and their coexistence to be examined.

Table 7 places our quantitative results in the context of previous web tracking measurement studies. Direct comparisons must be interpreted cautiously because the studies differ in dataset size, website ranking, measurement period, crawling methodology, and the specific mechanisms reported. Some works report aggregate fingerprinting prevalence [7, 30, 46], whereas others provide results for individual APIs or tracking mechanisms [10, 17, 21, 27, 49]. Nevertheless, the comparison provides useful context for assessing how the prevalence observed in our study relates to earlier measurements.

The results also show that tracking prevalence and coexistence vary across website popularity ranges. However, these descriptive differences should be interpreted cautiously. Confidence intervals overlap for several mechanisms, and the statistical analyses do not provide sufficient evidence to conclude that multilayered tracking is associated with a particular popularity range. The following subsections therefore discuss the observed patterns primarily in relation to the different tracking mechanisms and their coexistence, while explicitly considering the methodological differences with previous studies.

7.1 From isolated mechanisms to multilayered tracking ecosystems

One of the main findings of the joint analysis is the frequent coexistence of multiple tracking mechanisms. Only 13.11% of the analyzed websites showed none of the four tracking categories considered in the study, whereas 61.48% exhibited at least three categories: 45.36% exhibited exactly three and 16.12% exhibited all four. The most frequent specific combination was web analytics, cookies, and web storage, observed on 30.87% of websites, while all four categories, including active browser fingerprinting, coexisted on 16.12%.

These findings support a view of modern web tracking as a multilayered ecosystem rather than one dominated by a single mechanism. The coexistence of stateful mechanisms such as cookies and browser storage with stateless fingerprinting and analytics-related technologies indicates that privacy protections focused on only one tracking vector may leave other identification or measurement mechanisms unaffected. This observation is particularly relevant as browsers and privacy tools increasingly restrict traditional mechanisms such as third-party cookies [17, 33, 40]. This multilayered pattern is also consistent with previous studies suggesting that restrictions on third-party cookies are contributing to the diversification of tracking strategies rather than eliminating tracking altogether [11, 14, 27].

Although the descriptive results show differences in the prevalence of multilayered tracking across ranking intervals, the binomial logistic regression did not identify statistically significant differences according to website popularity. The overall model approached statistical significance ($\chi^2(2) = 5.67$, $p = 0.059$), but no individual ranking interval was statistically significant. Therefore, the observed coexistence of three or more tracking-mechanism categories cannot be attributed to a specific popularity range based on the present dataset.

7.2 Fingerprinting as a widely deployed tracking strategy

Active browser fingerprinting represents a substantial component of the tracking ecosystem observed in our measurements, being detected on 31.28% of the analyzed websites. This prevalence is higher than that reported in several earlier crawler-based studies, including Englehardt and Narayanan [17], FP-Inspector [30], and FPMon [21], and is consistent with more recent studies reporting an increasing presence of fingerprinting techniques [46, 59]. Direct quantitative comparisons should nevertheless be interpreted cautiously because these studies differ in measurement periods, datasets, crawling conditions, and fingerprinting detection criteria.

The distribution across individual fingerprinting techniques also differs from that emphasized in some earlier studies. JS Enumeration (25.55%) and WebGL (20.90%) are the most prevalent techniques in our measurements, whereas Canvas Font fingerprinting is considerably less common (0.96%). This pattern suggests that current fingerprinting implementations make substantial use of browser-exposed software, hardware, and rendering characteristics beyond the techniques traditionally emphasized in earlier measurements. WebRTC fingerprinting is also observed in our dataset and appears more frequently than in some earlier large-scale studies [17], although methodological differences prevent interpreting this comparison as direct evidence of temporal growth.

At the same time, the prevalence measured by an automated crawl depends strongly on the browsing conditions under which fingerprinting behavior is triggered. Annamalai et al. [46], for example, reported substantially higher Canvas fingerprinting prevalence than that observed in our crawler-based measurement. This difference is consistent with the limitation discussed in Section 4.1: fingerprinting behaviors that are activated only after actions such as scrolling, logging in, or interacting with consent interfaces may remain unobserved during automated landing-page crawls.

These observations reinforce the importance of considering interaction-dependent behavior when interpreting crawler-based fingerprinting measurements. As reported by

Table 7 Comparison of studies analyzing tracking, fingerprinting, storage, and analytics technologies

	Studies										
Metric/study	[17]	[7]	[30]	[21]	[49]	[10]	[2]	[37]	[27]	[46]	Ours
General information											
Year	2016	2010–2019	2019	2020	2021	2023	2023	2024	2024	2024	2024
Sites	1M	100K	100K	10K	50	100K	10K	10K	50	3K	1K
Failed (%)	8.27	43.09[a]	11.26	8.08	0	19.38	5.53	20.6	12	0.5	26.8
Ranking	Alexa	Alexa	Alexa	Alexa	Alexa	Tranco	Tranco	Tranco	Alexa	CrUX[j]	Tranco
Cookies											
Cookies (%)	56	NA	NA	NA	NA	NA	NA	39[b]	NA	NA	82.65
1st-party cookies (%)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	81.28
3rd-party cookies (%)	NA	NA	NA	NA	NA	NA	NA	NA	36.36	NA	57.38
Fingerprinting techniques											
Fingerprinting (%)	≤2[a]	1.16/3.70[c]	10.18	19[g]	14-30[e]	8–69[h]	NA	NA	29.54	15.7[i]	31.28
FP top-1K (%)	NA	NA	30.6	NA	NA	NA	NA	NA	NA	NA	31.28
Canvas (%)	1.6	NA	NA	17.85	10	7.74	NA	NA	19	NA	7.92
Canvas top-1K (%)	5.1	NA	NA	NA	NA	NA	NA	NA	NA	NA	7.92
Canvas Font (%)	0.32	NA	NA	14.79	NA	NA	NA	NA	NA	NA	0.96
Canvas Font top-1K (%)	2.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.96
WebRTC (%)	0.07	NA	NA	NA	6	NA	NA	NA	3	NA	2.73
WebGL (%)	NA	NA	NA	NA	14	NA	NA	NA	14	NA	20.9
AudioContext (%)	0.05	NA	NA	NA	NA	NA	NA	NA	NA	NA	3.01
JS Enumeration (%)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	25.55
Storage											
Web storage (%)	NA	NA	NA	NA	NA	NA	0.54	NA	≤61[f]	NA	78.96
Local storage (%)	NA	NA	NA	NA	44.1	NA	73.18[d] /39[e]	NA	39	NA	76.5
Session storage (%)	NA	NA	NA	NA	38	NA	25.44[d] /14[e]	NA	22	NA	70.36
IndexedDB (%)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	27.6
Web analytics											
Google Analytics (%)	NA	NA	NA	NA	40	47.9	NA	NA	NA	NA	50.82
Facebook Pixel (%)	NA	NA	NA	NA	20	NA	NA	NA	3	NA	8.8
Microsoft UET (%)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	3.01
Hotjar (%)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	3.69

All values are percentages unless otherwise indicated. NA = Not Available.
[a] Script prevalence instead of website prevalence.
[b] Percentage of websites tracking users after cookie rejection.
[c] Values correspond to 2016/2018, respectively.
[d] Flow distribution instead of website prevalence.
[e] Estimated value.
[f] Estimated from the data reported by the authors.
[g] Intensive fingerprinting only (classified as "High").
[h] Range between "Very High" and "High + Very High" categories.
[i] Based on real user interaction instead of automated crawling.
[j] Chrome User Experience Report

Annamalai et al. [46], fingerprinting scripts may be activated only after specific user interactions, meaning that measurements restricted to initial page loading can underestimate both prevalence and technique diversity.

Fingerprinting also frequently coexists with cookies, browser storage, and web analytics in our dataset. This reinforces the multilayered pattern discussed in the previous subsection and suggests that fingerprinting should be considered alongside, rather than independently of, stateful tracking mechanisms when characterizing modern tracking deployments.

The prevalence and coexistence of fingerprinting observed in this study also highlight the limitations of protections focused exclusively on stateful tracking mechanisms such as third-party cookies. Mitigating fingerprinting requires complementary protections at the browser and API level, particularly because fingerprinting can operate without relying on persistent client-side identifiers.

Future longitudinal measurements will be necessary to determine how the prevalence, diversity, and activation conditions of browser fingerprinting evolve as browsers introduce additional privacy protections and websites adapt their tracking strategies.

7.3 Fingerprinting and cookies under privacy laws

Stateful tracking mechanisms also remain highly prevalent in our measurements. First-party cookies were observed on 81.28% of websites and third-party cookies on 57.38%, while Local Storage and Session Storage were observed on 76.50% and 70.36%, respectively. IndexedDB was less prevalent, at 27.60%. These results show that restrictions affecting traditional third-party cookies coexist with extensive use of first-party cookies and alternative client-side storage mechanisms. Previous work has similarly documented the increasing relevance of first-party cookies and browser storage in contemporary tracking ecosystems [2, 11, 27].

Our cookie measurements are also higher than those reported in some previous studies. Englehardt and Narayanan [17] reported cookies on 56% of websites, whereas Lin et al. [37] reported 39% in a study focused on cookie tracking after consent rejection. These values are not directly comparable because the studies differ in measurement conditions, scope, and cookie-detection criteria. Nevertheless, together with previous work describing increasing reliance on first-party or server-assisted tracking [11, 22], our results confirm that cookie-based and storage-based mechanisms remain important components of current tracking ecosystems.

The widespread use of cookies, fingerprinting techniques, and persistent browser storage raises important legal and privacy concerns. Under the European Union's General Data Protection Regulation (GDPR), online identifiers and other information that can identify or make a natural person identifiable may constitute personal data [20]. Consequently, cookies and browser-fingerprinting data may fall within the scope of data-protection requirements when they are used to identify, recognize, or profile users or devices. Their deployment may also be subject to consent and transparency requirements depending on the processing purpose and the applicable legal framework [8, 51].

Similarly, the California Consumer Privacy Act (CCPA) defines personal information broadly and encompasses identifiers and information related to online activity [8]. Other jurisdictions have adopted comparable comprehensive data-protection frameworks. The Brazilian General Data Protection Law (LGPD) regulates the processing of personal data [8], while India's Digital Personal Data Protection Act (DPDPA) establishes requirements for the processing of digital personal data, including consent-based processing in applicable cases [6]. Proposed U.S. frameworks such as the American Data Privacy and Protection Act (ADPPA) have likewise defined covered data broadly as information that identifies or is reasonably linkable to an individual or device [5]. Although these frameworks differ in their specific requirements, they illustrate the broader regulatory attention given to online identifiers, behavioral data, and tracking-related processing.

Despite this regulatory attention, fingerprinting remains comparatively difficult for users to detect and control and may operate without meaningful user awareness or consent [33, 35]. Unlike cookies, which are commonly exposed through browser settings and consent interfaces, fingerprinting scripts may be less visible to users and are technically more difficult to detect or prevent [17, 33, 35, 52]. Our measurements detected active browser fingerprinting on 31.28% of the analyzed websites, and the coexistence analysis shows that fingerprinting frequently appears together with cookies and web storage. This coexistence is relevant from a compliance perspective because controls focused only on cookies may not provide users with equivalent visibility or control over alternative identification mechanisms.

These findings are consistent with prior research questioning whether consent interfaces provide effective control over the full range of tracking mechanisms [8, 26, 51]. Studies of cookie banners and browsing conditions have shown that consent-interface design and user interaction can substantially affect observed tracking behavior, while alternative mechanisms such as fingerprinting and persistent client-side storage may remain outside cookie-centric controls [8, 24, 48, 58].

Taken together, these observations highlight a remaining gap between regulatory requirements and the technical mechanisms available to users for understanding and controlling tracking. Addressing this gap may require clearer regulatory guidance, effective enforcement, and browser-level protections that account for multiple tracking mechanisms rather than focusing exclusively on cookies. Similar concerns regarding the difficulty of controlling browser fingerprinting and other tracking mechanisms have been raised in prior work [3, 33, 35].

7.4 The limits of current web privacy measurement approaches

The differences observed between our results and those reported in previous studies also highlight the methodological limitations of current web privacy measurement approaches. These differences should not necessarily be interpreted as contradictions, since different measurement methodologies, browsing conditions, and detection criteria may capture different subsets of the tracking behaviors deployed on websites.

Crawler-based approaches such as OpenWPM provide scalable, reproducible, and low-cost measurements suitable for longitudinal and large-scale web privacy studies. However, automated crawlers may fail to observe fingerprinting behaviors encountered during real-user browsing. Annamalai et al. [46], for example, showed that 45% of the fingerprinting websites observed during real-user browsing sessions were entirely missed by equivalent automated crawls. The authors attribute this discrepancy mainly to crawlers being unable to access authentication-protected pages, being blocked by bot-detection mechanisms, and failing to trigger fingerprinting scripts activated by specific user interactions. This difference in measurement conditions can lead real-user studies to observe fingerprinting behaviors that remain absent from crawler-based datasets.

Conversely, specialized approaches such as FP-Inspector [30] and FP-Tracer [10] are specifically designed to improve fingerprinting detection through techniques such as semantic analysis or dynamic taint tracking. Their objective, however, differs from broader measurement frameworks intended to jointly analyze multiple tracking mechanisms. As also illustrated by the tool-level comparison in Table 1, privacy measurement tools differ substantially in the mechanisms they support and in the depth with which individual tracking techniques are analyzed.

Taken together, these observations indicate that current privacy measurement approaches involve trade-offs between measurement breadth, detection depth, scalability, and the ability to reproduce real-user browsing conditions. Consequently, no single methodology provides complete visibility into all forms of tracking behavior considered across the literature.

Future web privacy measurements could therefore benefit from combining complementary approaches, such as scalable browser instrumentation, controlled user-interaction simulation, real-user telemetry, and specialized fingerprinting analysis. Such combinations could provide broader coverage while preserving the reproducibility required for longitudinal measurement studies.

7.5 Limitations of this work

In addition to limitations inherent to crawler-based web measurement approaches discussed above, our study presents several specific limitations. First, OpenWPM may be detected or blocked by websites, as discussed by Krumnow et al. [32]. Together with the absence of simulated user interactions, this may prevent some tracking behaviors from being observed and therefore lead to an underestimation of their presence. In addition, passive fingerprinting is outside the scope of the study because it relies on information processed on the server side and cannot be directly observed through the client-side instrumentation employed by OpenWPM.

A second limitation concerns the interpretation of cookies and browser storage. The current analysis detects their presence and activity but does not determine whether individual cookies or stored values are used for tracking or for legitimate functional purposes. Distinguishing between these cases may require analyzing stored identifiers and their evolution across browsing contexts and, in some cases, triggering additional website behavior. Consequently, the prevalence values reported for cookies and web storage should not be interpreted as equivalent to the prevalence of tracking-oriented use.

Web analytics detection also depends on known domains and request patterns associated with the four platforms considered in the study. Undocumented endpoints, deployment-specific configurations, or future changes in these services may therefore remain undetected.

The final analysis was conducted on 732 of the 1,000 domains included in the selected Tranco list. Although many unsuccessful crawls corresponded to infrastructure-oriented services rather than conventional user-facing websites, the 26.8% crawl failure rate may still introduce sampling bias and should be considered when generalizing the results.

Finally, the measurement pipeline is necessarily dependent on the tracking techniques, browser APIs, domains, and heuristic thresholds considered at the time of the study. New tracking mechanisms, changes in browser APIs, or modifications in tracker behavior may reduce the effectiveness of the current detection criteria over time. The results should therefore be interpreted as a temporal snapshot of the tracking ecosystem under the measurement configuration used in this study, and the instrumentation and detection rules may require periodic updates to preserve their coverage.

8 Conclusions and future work

This study investigated the prevalence and coexistence of multiple stateful and stateless tracking mechanisms across highly visited websites. To address the limited availability of detailed measurements jointly examining these mecha-

nisms, we applied an OpenWPM-based measurement and analysis pipeline to the Tranco Top-1000 websites, successfully crawling 732 domains. The analysis jointly considered cookies, web storage, web analytics platforms, and active browser fingerprinting.

From a technical perspective, the work extends two complementary components of the OpenWPM-based measurement and analysis pipeline. OpenWPM's JavaScript instrumentation was extended to collect the browser-side information required for the tracking mechanisms considered in the study, while OpenWPM-Data-Analytics was enhanced to support their subsequent detection, classification, and quantification. The resulting pipeline was validated with a sensitivity of 94.55%, specificity and precision of 100%, and an overall accuracy of 97.50% on the validation set containing known tracking mechanisms.

From an empirical perspective, cookies and web storage were the most prevalent mechanisms, being observed on 82.65% and 78.96% of the successfully crawled websites, respectively, while active browser fingerprinting was detected on 31.28%. More importantly, 61.48% of the websites exhibited at least three of the four mechanism categories considered in the study. These results show that modern web tracking is frequently multilayered, combining stateful and stateless mechanisms within the same websites.

Although descriptive differences were observed across popularity ranking intervals, the statistical analysis did not provide sufficient evidence to associate multilayered tracking with a particular ranking interval.

Overall, the findings reinforce the importance of evaluating multiple tracking mechanisms jointly, since measurements focused on a single technology provide only partial visibility into websites where cookies, browser storage, analytics, and fingerprinting coexist.

Future technical work will focus on improving measurement coverage and robustness. This includes investigating less detectable JavaScript instrumentation, extending the analysis of cookies and browser storage beyond presence and access counts, incorporating additional fingerprinting mechanisms and web analytics platforms, and improving the scalability of OpenWPM-Data-Analytics through parallel processing.

Future research should also include longitudinal measurements to assess how tracking prevalence and coexistence evolve over time, as well as crawling scenarios incorporating controlled user interactions and geographical variation. A common validation framework for privacy measurement tools would further facilitate systematic comparisons of their coverage, accuracy, and methodological trade-offs.

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Data Availability Data and source code are available at https://github.com/fernandojavier-almenaras-um-es/OpenWPM-Web-Analysis.

Declarations

Competing interests The authors declare no competing interests.

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